

Iterative Methods in Matrix Algebra - Summary

Part 1: The “Rulers” (Norms)

Vector Norms: For a vector $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$:

- l_∞ Norm: Maximum absolute component.

$$\|\mathbf{x}\|_\infty = \max_{1 \leq i \leq n} |x_i|$$

- l_2 Norm: Euclidean length.

$$\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$$

Matrix Norms: For $A \in \mathbb{R}^{n \times n}$:

- l_∞ Norm (row sum):

$$\|A\|_\infty = \max_{1 \leq i \leq n} \sum_{j=1}^n |a_{ij}|$$

- l_1 Norm (column sum):

$$\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^n |a_{ij}|$$

Part 2: The “Law of Convergence” (Spectral Radius ρ)

- Spectral radius of a matrix T :

$$\rho(T) = \max |\lambda_i|, \quad \lambda_i \text{ eigenvalues of } T$$

- Iterative method converges iff $\rho(T) < 1$.

- **Iteration matrix formulas:**

- Jacobi:

$$T_J = D^{-1}(L + U)$$

- Gauss-Seidel:

$$T_{GS} = (D - L)^{-1}U$$

Here $A = D - L - U$ (next part) is used to construct T .

Part 3: Matrix Decomposition ($A = D - L - U$)

Split A into three parts:

- D : Diagonal matrix, $d_{ii} = a_{ii}$
- L : Strictly lower triangular, $l_{ij} = -a_{ij}$ for $i > j$
- U : Strictly upper triangular, $u_{ij} = -a_{ij}$ for $i < j$

Example:

$$A = \begin{bmatrix} 5 & -1 \\ -2 & 8 \end{bmatrix}, \quad D = \begin{bmatrix} 5 & 0 \\ 0 & 8 \end{bmatrix}, \quad L = \begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix}, \quad U = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Part 4: Iterative Methods

1. Jacobi Method:

$$x_i^{(k)} = \frac{1}{a_{ii}} \left[b_i - \sum_{j \neq i} a_{ij} x_j^{(k-1)} \right]$$

Matrix form:

$$\mathbf{x}^{(k)} = T_J \mathbf{x}^{(k-1)} + D^{-1} \mathbf{b}$$

2. Gauss-Seidel Method:

$$x_i^{(k)} = \frac{1}{a_{ii}} \left[b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(k)} - \sum_{j=i+1}^n a_{ij} x_j^{(k-1)} \right]$$

Matrix form:

$$\mathbf{x}^{(k)} = T_{GS} \mathbf{x}^{(k-1)} + (D - L)^{-1} \mathbf{b}$$

! The matrices T_J and T_{GS} are the **iteration matrices**, which are used to compute the spectral radius for convergence checking.

Part 5: The SDD Shortcut

- Strictly Diagonally Dominant (SDD):

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}| \quad \forall i$$

- If A is SDD, **Jacobi and Gauss-Seidel converge for any initial guess**. - No need to calculate eigenvalues; convergence is guaranteed.